

Blunt Assault: 'Million Dollar Baby'

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Contrasting reports exist in the literature regarding the management of patients with blunt assault to the head, neck, and face and more importantly for clearing the cervical spine. The purpose of our study was to determine the risk of cervical spine injuries after blunt assault to the head, neck, and face and its association with other injuries. We performed a retrospective case review of all blunt assault trauma admissions to the head, neck, and face at our Level I trauma center. We identified all patients who were assaulted with hands and feet and blunt instruments. A total of 3286 patients with blunt assault to the head, neck, and face were identified of whom 11 (0.003%) were found to have a cervical spine fracture or cervical spine subluxation. None of the patients had a cervical spinal cord injury. The 11 patients composed our study population with a mean age of 39 ± 7.8 years, 100 per cent were male, and the mean Injury Severity Score was 12 ± 7.9 . Five (45%) patients required surgery for stabilization of the cervical spine. Mortality was reported in only one patient who had a C7 transverse process fracture. Cervical spine injury after blunt assault is rare but does occur and encompasses significant injuries requiring surgical intervention. However, these injuries are the result of direct blows to the cervical spine and we suggest that assaulted patients with no direct trauma to the neck do not require an exhaustive evaluation of the cervical spine.

IN OUR EVERYDAY practice, we see a large number of patients in the emergency department with blunt head, neck, or facial trauma. The Advance Trauma Life Support course teaches to assume a cervical spine injury in patients with blunt injury to the head, face, and neck and mandates an elaborate cervical spine examination.¹ Multiple contrasting reports exist in the literature for the management of these patients with blunt assault to the head, neck, and face and more importantly for clearing the cervical spine.^{2, 3} However, most of these studies have failed to distinguish between the various mechanisms of blunt assault with most of their patient group composed of motor vehicular injuries. The pattern of cervical spine injury occurring in a patient after a blunt assault varies depending on the mechanism of the assault. Motor vehicular accidents cause a flexion-extension pattern of cervical injury, whereas a direct blow to the neck causes an impact injury.⁴ A direct blow to the neck produces more serious injuries compared with an indirect force to the head, neck, or the face as a result of the difference in kinematics of force.⁵ The primary aim of our study was to determine the risk of cervical spine injuries after blunt assault to the head, neck,

and face and its association with other injuries. Secondly, we wanted to determine whether we are burdening our trauma centers with the doctrine of complete cervical spine evaluation for all blunt assault cases to the head, neck, and face.

Methods

We performed a 14-year retrospective case review of all blunt assault trauma admissions to the head, neck, and face presenting to our academic Level I trauma center. All patients who were assaulted with hands, feet, and blunt instruments were identified through our trauma registry using E codes (973, 968.2, 960, 805.0, 805.1, 806.0, 806.1, and 952). Patients with a cervical spine injury or a cervical column injury were identified and a chart review of these patients was performed.

Patient variables were collected using Microsoft Excel 2003 (Microsoft Corporation, Redmond, WA), which included demographics, mechanism of injury (hands and feet vs blunt object), injury sustained identified by *International Classification of Diseases, 9th Revision* codes, admission vitals, Glasgow Coma Scale (GCS), Abbreviated Injury Score, Injury Severity Score (ISS), and outcomes. Statistical analysis was performed using SAS System, Version 8.2 (SAS Institute Inc., Cary, NC).

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This study was approved by the Institutional Review Board and adhered to established guidelines on the treatment of human subjects.

Results

During the 14-year study period, 3286 patients with blunt assault to the head, face, and neck were identified of whom 2286 (70.0%) had both head and/or neck and/or face injuries. Eleven (0.003%) patients were found to have a cervical spine fracture or cervical spine subluxation (CSFS). None of these patients had a cervical spinal cord injury. In these patients, the mean age was 39 ± 7.8 years, 100 per cent were male, and the mean ISS was 12 ± 7.9 (Table 1). The details of these 11 patients and their outcomes are demonstrated in Tables 2 and 3, respectively.

Case 1 presented with facial fractures and left upper extremity radiculopathy sustained from being punched

and kicked. Cervical spine imaging revealed acute and chronic changes including a C5–C6 and C6–C7 compression fracture, C3–C4 mild degenerative changes, C4–C5 mild discogenic osteophytes, and C5–C6 mild left neuroforaminal narrowing. The mechanism of assault could not be recalled by Case 2. Radiographic findings demonstrated a C3 anteroinferior corner “teardrop fracture,” prevertebral soft tissue fluid collection, and no ligamentous injury. An associated subdural hemorrhage was also noted. Neither of these two patients required surgical intervention for treatment of cervical spine injuries.

Four patients were assaulted with a baseball bat. On admission, Case 3 had a GCS of 7 and was intubated immediately in the emergency department. Imaging of the neck and head revealed anterior subluxation of the C4 vertebra, an associated depressed parieto-occipital skull fracture, and a cerebral contusion. On neurologic examination, right hemiparesis was documented resulting from the cerebral injury. The patient was treated with halo vest immobilization for the cervical spine injury. Case 4 was found to have a C1–C2 subluxation with no other significant injuries. The patient was treated with open reduction and internal fixation (ORIF) of C1–C2 using bone graft (Brooks fusion) and halo vest immobilization. A C1 ligamentous injury and an open non-depressed occipital fracture was identified in Case 5, which were treated with halo vest immobilization. In Case 6, degenerative joint changes of C5–C6 with questionable fracture and anterior compression at C5–C6 were found. Additionally, an associated basilar skull fracture, facial

TABLE 1. Admission Characteristics

Variable	Blunt		
	All (n = 11)	Instrument Hands and Feet (n = 9)	(n = 2)
Age (years)	39 ± 7.8	36 ± 6.5	49 ± 2.8
Admission SBP	132 ± 24.3	128 ± 21.0	151 ± 38.9
Admission HR	101 ± 21.3	104 ± 21.9	84 ± 5.7
Admission RR	18 ± 6.2	17 ± 6.8	20 ± 0
GCS	13 ± 3.8	13 ± 4.1	15 ± 0
ISS	12 ± 7.9	11 ± 8.5	16 ± 3.5

SBP, systolic blood pressure; HR, heart rate; RR, respiratory rate; GCS, Glasgow Coma Scale; ISS, Injury Severity Score.

TABLE 2. Patient Injury Characteristics

Case No.	Mechanism	C-spine Injuries	Signs/Symptoms	Procedures
1	Fists and feet	C5–C7 compression fracture	Left upper extremity radiculopathy	None
2	Unclear	C3 anteroinferior corner fracture; “teardrop fracture”; prevertebral soft tissue fluid collection		
3	B-B	C4–C5 anterior subluxation	Right hemiparesis	Halo
4	B-B	C1–C2 subluxation	C1–3 tenderness	ORIF Brooks fusion of C1–C2 using bone graft; halo
5	B-B	C1 ligamentous injury	None	Halo
6	B-B	C5–C6 anterior fracture		C-collar
7	Metal bar	Stable C5–C6, C6–C7 laminar fractures; left C6 pars fracture; C7 transverse process fracture	None	
8	B-I	C6–C7 subluxation “locked facets”	Left upper extremity radiculopathy	ORIF C6/7 and bone graft; halo
9	B-I	C3–C4, C4–C5, C5–C6 disk bulges		
10	Blind man’s cane	C7 right transverse process fracture		
11	Punched by partner and hit head against cement parking block	C4–C5 jumped facet	C5 radiculopathy	ORIF and bone graft

B-B, baseball bat; B-I, blunt instrument; ORIF, open reduction and internal fixation.

TABLE 3. *Outcomes*

Variable	All (n = 11)	Blunt Instrument (n = 8)	Hands and Feet (n = 2)
Died, no. (%)	1 (9.1)	1 (8.3)	0
ICU admission, no. (%)	5 (45.5)	4 (44.4)	1 (50)
ICU LOS (days)	8 ± 10.6	9 ± 12.1	5 ± 0
Hospital LOS (days)	12 ± 11.6	12 ± 12.8	9 ± 4.9

ICU, intensive care unit; LOS, length of stay.

fractures, and a radial fracture were also documented. The patient was treated with a cervical collar.

Four patients were assaulted with other blunt instruments, namely a metal pipe, stick, and cane. Case 7 sustained stable C5–C6 and C6–C7 laminar fracture, a left C6 pars fracture, and a C7 transverse process fracture. No treatment of the cervical spine was required. Case 8 presented with left upper extremity radiculopathy and a C6–C7 subluxation with locked facets. This patient underwent ORIF and bone graft of C6–C7 and halo vest immobilization. Case 9 was intubated in the trauma bay for a low GCS. Imaging revealed disk bulges at C3–C4, C4–C5, and C5–C6 without evidence of spinal cord impingement or ligamentous injury. Additionally, a skull fracture, an intraparenchymal brain hemorrhage, a lumbar spine fracture, and a hemopneumothorax were noted. The cervical spine did not require treatment. Case 10 was assaulted with a cane and sustained a right transverse process fracture of C7 and associated subdural and subarachnoid hemorrhages. This patient died from a cardiopulmonary arrest in the intensive care unit on Day 4 after multiple complications.

Case 11 sustained a punch and fell and struck his neck against a cement parking block. He presented with a C-5 radiculopathy and was found to have a jumped facet of the C4–C5 vertebrae and underwent ORIF and bone graft fixation.

DISCUSSION

In our study, we demonstrated that blunt traumatic assault with hands, feet, or a blunt object can be a significant mechanism of injury for development of cervical spine injury and other associated injuries. In our study, three of the four patients who were assaulted with a baseball bat required surgical intervention. One patient required an ORIF with bone graft and halo vest immobilization, two patients required halo vest immobilization, whereas one patient was treated with a cervical collar. Of the two patients who were punched and kicked, one did not require treatment of cervical spine injuries, although he did present with a left upper extremity radiculopathy associated with chronic cervical vertebral disk changes. The second patient was assaulted with a fist and fell, striking his occiput against a

cement parking block. This patient presented with a C5 radiculopathy and required ORIF with bone graft for stabilization but did not have a cervical spinal cord injury. Three patients were battered with unknown instruments of whom; one patient presented with a left upper extremity radiculopathy and underwent ORIF with bone graft and halo vest immobilization; the other two did not require treatment of cervical spine injuries. The patient with an undetermined mechanism of assault did not require any therapeutic intervention for the cervical spine, whereas the patient who was struck with a metal pipe had multiple stable fractures of his cervical spine and was discharged without any treatment for these injuries. There was only a single reported mortality in a patient who sustained a C7 transverse process fracture, which did not require treatment. This patient had extensive associated injuries including subdural and subarachnoid hemorrhage and he developed multiple organ failure, which ultimately led to cardiopulmonary arrest and death.

Our study reproduces the finding of multiple papers that have found a very low incidence of cervical spine injury from blunt assault.^{2, 3, 5, 6} A study by Patton et al. of 102 patients with blunt assault to the head and neck found only one patient with a stable spinous process fracture, who was discharged the next day. They found no clinically significant cervical spine injuries and concluded that victims of blunt assault, regardless of neurological status, can be cleared on the basis of plain radiographic evidence alone.⁷ In contrast, our study shows that there were five patients requiring surgical treatment after their beating (two patients had an ORIF, bone graft, and halo; two had a halo; and one had an ORIF and bone graft only). In addition, two of the patients presented with neurological findings of upper extremity radiculopathy.

It is intuitive that blows from a blunt instrument like a baseball bat or a metal pipe will produce injuries of greater severity than punches and kicks. Studies have demonstrated that a baseball bat when swung full swing can produce enough force to crush a human skull,⁸ whereas kicks or punches produce comparably lesser forces.^{9, 10} Surprisingly, a study by Shepherd et al. reported a significantly higher incidence of fractures in attacks with fists and feet compared with attacks with blunt weapons. They attributed this finding to the greater momentum of fists and feet as compared with most blunt objects.¹¹ In our study, we found that patients assaulted with a blunt object sustained serious injuries requiring surgical management of the cervical spine compared with patients assaulted with punches and kicks. All patients in our current series sustained CSFS from direct blows to their necks. According to Walicko et al., a greater force is required to cause injury a cervical spine than that conveyed by a punch.⁹ One patient

Associated Injury with Cervical Spine Injury after Blunt Assault

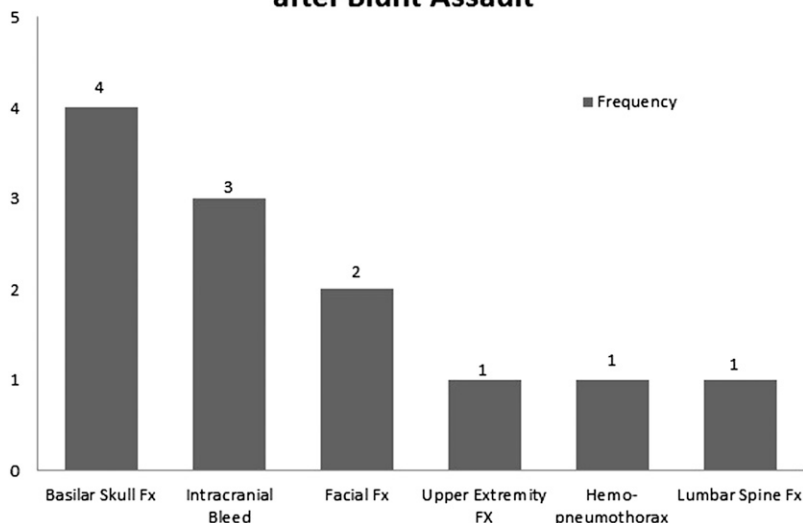


FIG. 1. Associated injuries with cervical injury after blunt assault.

in our series required surgical management of a cervical spine injury after a punch. However, this is our “million dollar baby” who sustained a CSFS after falling and striking his head against a cement parking block, not directly from a fist. Furthermore, we found that a baseball bat can cause severe cervical spine injuries (three patients required a halo), unlike the findings from Baltimore of 75 patients who did not have any acute injuries to their cervical spine.⁸

In our study, we found that head injuries such as skull fracture and intracranial bleeds are often associated with cervical spine injuries (Fig. 1). However, it is highly unlikely that a single blow to the head or face from a single punch, kick, or blow from a blunt weapon causes both an intracranial injury and a CSFS. As described earlier, a directed blow to a localized region is most often the cause of the injury. Thus, we believe that these patients were assaulted multiple times and hence sustained multiple injuries. Our findings are similar to the ones previously reported by John et al. and Shepard et al.^{3, 11}

Conclusions

Although the incidence of cervical spine fracture or cervical spine subluxation from blunt assault is exceedingly rare, and no injuries to the cervical spinal cord were identified, a small percentage of patients sustained significant cervical spinal column injuries requiring therapy, as shown in our 11 patients. It is apparent that the transmission of biomechanical forces to the cervical spine varies widely among blunt mechanisms such as motor vehicle collisions, fall from a height, and assault. Prospective studies are needed to establish guidelines for a more directed evaluation of

the cervical spine in patients sustaining blunt assault. As described earlier, it is the direct blows to the neck and not the face and head that result in significant cervical spine injuries and we suggest that patients with this mechanism of injury warrant a more thorough evaluation of their cervical spine.

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